

POLITECNICO
MILANO 1863

SYSTEMS AND METHODS FOR BIG AND UNSTRUCTURED DATA

Project Work 2022

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Project Work

Group work
Incremental deliveries

Course Evaluation

30% group project delivery

20% closed-book, multi-choice questionnaire

50% written exam, open questions and exercises

Project Work

Groups of max. 5 people

Selection of one project topic

Data and architecture design and implementation

Activity

Specification of problem
Data design
Graph
Document
Spark

Topic Description

Bibliography Database

Scientific articles
Authors, affiliations
Journals and conferences
DOI

Implementation: Neo4J

Scientific Papers

Full description of document structure and content

- Title, Abstract, Authors (with affiliations, email, bio)
- Metadata (keywords)
- Publication details (journal, volume, number, date, pages)
- Sections: title, text (by paragraph), subsections, figures (image URL and caption)
- Bibliography (set of refs.)

Implementation: MongoDB

References

Dblp.org

full XML dump: https://dblp.uni-trier.de/xml/

DOI.org

ORCID.org

Datasets

(examples of sources for possible importing)

https://github.com/niderhoff/nlp-datasets

https://imerit.net/blog/25-best-nlp-datasets-for-machine-learning-all-pbm/

https://odsc.medium.com/20-open-datasets-for-natural-language-processing-538fbfaf8e38

(example of sensible random generators)
https://www.mockaroo.com/

Deliveries

Delivery #1

ER model
Graph DB
Data upload (importing)
Data queries

Cypher Queries for Delivery #1

Minimum 5 (diverse) data creation/update commands
Minimum 10 queries
Minimum complexity of queries:

- 3 nodes, conditions
- 3 nodes, conditions, aggregation
- 5 nodes, conditions, aggregations, limits
- Functions (minimum path)

Check complexity / performance time

Delivery #2

MongoDB implementation of the collection
Data upload (importing)
Data queries

MongoDB Queries for Delivery #2

Minimum 5 (diverse) data creation/update commands
Minimum 10 queries
Minimum complexity of queries:

- Filtering conditions
- Filtering conditions, aggregation
- Filtering conditions (also on internal subdocs)
- Filtering through unwind
- Filtering through refs

Check complexity / performance time

THANKS!
QUESTIONS?



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